

Justify your answers and show your work.

1. $\int \sin^2 3x \, dx$

2. $\int \cos^3 x \, dx$

3. $\int \sin^2 x \cos^2 x \, dx$

4. $\int \cos^3 x \sin x \, dx$

5. $\int \frac{dx}{\sqrt{4-x^2}}$

6. $\int_0^4 \sqrt{16-x^2} \, dx$

7. $\int \sqrt{16-x^2} \, dx.$

8. $\int \frac{3x+11}{x^2-x-6} \, dx$